Question 58.

Use RK4 method with $h = 0.1, 0.05, 0.025$ to estimate $y(2)$ for the IVP

$$y'(t) = \frac{2y}{t} + t^2 e^t, \quad y(1) = 0$$

which has the exact solution $y = t^2(e^t - e)$. Find how the numerical error varies with h.

Question 59.

Consider RK4 applied to a general initial value problem of the form

$$y'(t) = f(t) \quad \text{with } y(t_0) = y_0.$$

In other words, there is no dependence of f on y. Show that in this case the classical fourth-order Runge-Kutta method corresponds to Simpson's quadrature rule.

Computational Comparison of Runge-Kutta methods

The main computational effort in applying the Runge-Kutta methods is the evaluation of f. In the second-order methods, the local truncation error is $O\left(h^3\right)$, and the cost is two function evaluations per step. The Runge-Kutta method of order four requires 4 evaluations per step, and the local truncation error is $O\left(h^5\right)$. Butcher has established the relationship between the number of evaluations per step and the order of the local truncation error shown in table 4.1. This table indicates why the methods of order less than five with smaller step size are used in preference to the higher-order methods using a larger step size.

Evaluations per step	2	3	4	$5 \le n \le 7$	$8 \le n \le 9$	$10 \le n$
Best possible local truncation error	$O\left(h^3\right)$	$O\left(h^4\right)$	$O\left(h^5\right)$	$O\left(h^n\right)$	$O\left(h^{n-1}\right)$	$O\left(h^{n-2}\right)$

Table 4.1

One measure of comparing the lower-order Runge-Kutta methods can be described as follows - the Runge-Kutta method of order four requires four evaluations per step, whereas Euler's method requires only one evaluation. Hence if the Runge-Kutta method of order four is to be superior it should give more accurate answers than Euler's method with one-fourth the step size. Similarly, if the Runge-Kutta method of order four is to be superior to the second-order Runge-Kutta methods, which require two evaluations per step, it should give more accuracy with step size h than a second-order method with step size $h/2$.

Question 60.

Approximate the value of $y(1)$ for the IVP

$$y' = y - t^2 + 1, \quad 0 \le t \le 1, \quad y(0) = 0.5$$

using Euler's method with $h = 0.025$, the Midpoint method with $h = 0.05$, and the Runge-Kutta fourth-order method with $h = 0.1$. Solve the ODE and compare with the exact value of $y(1)$ to decide which method gives the most accurate result.

§4.3 Multistep Methods

All of the numerical methods that we have developed for solving initial value problems are one-step methods, because they only use information about the solution at time t_i to approximate the solution at time t_{i+1}. As i increases, that means that there are additional values of the solution, at previous times, that could be helpful, but are unused.

Multistep methods are time-stepping methods that do use this information. The main idea is to use the Newton-Cotes quadrature rule. A general multistep method has the form

$$\boxed{\sum_{j=0}^{s} \alpha_j y_{i+1-j} = h \sum_{j=0}^{s} \beta_j f\left(t_{i+1-j}, y_{i+1-j}\right)}$$

where s is the number of steps in the method ($s = 1$ for a one-step method), and h is the time step size, as before.

Note: By convention, $\alpha_0 = 1$, so that y_{i+1} can be conveniently expressed in terms of other values. If $\beta_0 = 0$, the multistep method is said to be explicit, because then y_{i+1} can be described using an explicit formula, whereas if $\beta_0 \neq 0$, the method is implicit, because then an equation, generally nonlinear, must be solved to compute y_{i+1}.

The main advantage of multistep methods is that only one new function evaluation is typically needed at each step. Most multistep methods fall into two broad categories:

- **Adams methods:**

  This method is named after its developer, British mathematician John Couch Adams, who also famously used mathematics to predict the existence and position of the planet Neptune.

  The general idea behind Adams methods is to find the interpolating polynomial through some number $s \geq 1$ of **past** nodes

  $$\left\{\left(t_{i-(s-1)}, f_{i-(s-1)}\right), \ldots, (t_{i-3}, f_{i-3}), (t_{i-2}, f_{i-2}), (t_{i-1}, f_{i-1}), (t_i, f_i), (t_{i+1}, f_{i+1})\right\}$$

  where $f_j = f(t_j, y_j)$, and then to integrate the interpolating polynomial over the interval $[t_i, t_{i+1}]$. This can be used to approximate $\displaystyle\int_{t_i}^{t_{i+1}} f(t, y(t))\, dt$.

  We can exclude the last node (t_{i+1}, f_{i+1}) if we want an explicit formula. In all Adams methods, $\alpha_0 = 1, \alpha_1 = -1$, and $\alpha_i = 0$ for $i = 2, \ldots, s$.

- **Backward differentiation formulas (BDF):**

  These methods also use polynomial interpolation, but for a different purpose - to approximate the derivative of y at t_{i+1}. This approximation is then equated to $f(t_{i+1}, y_{i+1})$. It follows that all methods based on BDFs are implicit, and they all satisfy $\beta_0 = 1$, with $\beta_i = 0$ for $i = 1, 2, \ldots, s$.

In our discussion, we will focus exclusively on Adams methods.

4.3.1 Explicit Adams Methods

Explicit Adams methods are called Adams-Bashforth methods. To derive an Adams-Bashforth method, we interpolate f at the points $t_i, t_{i-1}, \dots, t_{i-s+1}$ with a polynomial of degree $s-1$. We then integrate this polynomial exactly. It follows that the constants $\beta_j, j = 1, \dots, s$, are the integrals of the corresponding Lagrange polynomials from t_i to t_{i+1}, divided by h, because there is already a factor of h in the general multistep formula.

Question 61. □

Check that the global error of the resulting method is $\mathcal{O}(h^s)$.

Three-step Adams-Bashforth

Let's start with the equation

$$y_{i+1} = y_i + h[\beta_1 f(t_n, y_i) + \beta_2 f(t_{n-1}, y_{i-1}) + \beta_3 f(t_{n-2}, y_{i-2})].$$

Question 62. □

Explain why the constants β_j, $j = 1, 2, 3$ can be evaluated as

$$h\beta_j = \int_{t_i}^{t_{i+1}} \ell_j(t)\,\mathrm{d}t,$$

where $\ell_j(t)$ is the j-th Lagrange polynomial for the f-interpolant at t_i, t_{i-1} and t_{i-2}.

Now using a change of variable $u = \dfrac{t_{i+1} - t}{h}$, we can instead interpolate at the points $u = 1, 2, 3$, thus simplifying the integration. We conclude that the three-step Adams-Bashforth method is

$$y_{i+1} = y_i + \frac{h}{12}[23f(t_i, y_i) - 16f(t_{i-1}, y_{i-1}) + 5f(t_{i-2}, y_{i-2})]$$

Question 63. □

Derive the formula for the second order (for $s = 2$) Adams-Bashforth (AB2) method

$$y_{i+1} = y_i + \frac{h}{2}[3f(t_i, y_i) - f(t_{i-1}, y_{i-1})]$$

by integrating the Lagrange polynomials.

4.3.2 Implicit Adams Method

The same approach can be used to derive an implicit Adams method, which is known as an Adams-Moulton method. The only difference is that because t_{i+1} is an interpolation point, after the change of variable to u, the interpolation points $0, 1, 2, \dots, s$ are used. Because the resulting interpolating polynomial is of degree one greater than in the explicit case, the error in an s-step Adams-Moulton method is $\mathrm{O}\left(h^{s+1}\right)$, as opposed to $\mathrm{O}(h^s)$ for an s-step Adams-Bashforth method.

Question 64.

Check that the AB method for $s = 0$ and the AM method for $s = 1$ are simply the forward and the backward Euler methods respectively.

Using Predictor-Corrector Method with Adams-Moulton

An Adams-Moulton method can be impractical because, being implicit, it requires an iterative method for solving nonlinear equations, such as fixed-point iteration, and this method must be applied during every time step. An alternative is to pair an Adams-Bashforth method with an Adams-Moulton method to obtain an Adams-Moulton predictor-corrector method. Such a method proceeds as follows:

- **Predict:** Use the Adams-Bashforth method to compute a first approximation to y_{i+1}, which we denote by $\widetilde{y}_{i+1}$.
- **Evaluate:** Evaluate f at this value, computing $f(t_{i+1}, \widetilde{y}_{i+1})$.
- **Correct:** Use the Adams-Moulton method to compute y_{i+1}, but instead of solving an equation, use $f(t_{i+1}, \widetilde{y}_{i+1})$ in place of $f(t_{i+1}, y_{i+1})$ so that the Adams-Moulton method can be used as if it was an explicit method.
- **Evaluate:** Evaluate f at the newly computed value of y_{i+1}, computing $f(t_{i+1}, y_{i+1})$, to use during the next time step.

Example 4.3.31

We illustrate the predictor-corrector approach with the two-step Adams-Bashforth method

$$y_{i+1} = y_i + \frac{h}{2}[3f(t_i, y_i) - f(t_{i-1}, y_{i-1})]$$

and the two-step Adams-Moulton method

$$y_{i+1} = y_i + \frac{h}{12}[5f(t_{i+1}, y_{i+1}) + 8f(t_i, y_i) - f(t_{i-1}, y_{i-1})]$$

First, we apply the Adams-Bashforth method, and compute

$$\widetilde{y}_{i+1} = y_i + \frac{h}{2}[3f(t_i, y_i) - f(t_{i-1}, y_{i-1})].$$

Then, we compute $f(t_{i+1}, \widetilde{y}_{i+1})$ and apply the Adams-Moulton method, to compute

$$y_{i+1} = y_i + \frac{h}{12}[5f(t_{i+1}, \widetilde{y}_{i+1}) + 8f(t_i, y_i) - f(t_{i-1}, y_{i-1})].$$

This new value of y_{i+1} is used when evaluating $f(t_{i+1}, y_{i+1})$ during the next time step.

4.3.3 Drawbacks

One drawback of multistep methods is that because they rely on values of the solution from previous time steps, they cannot be used during the first time steps, because not enough values are available.

Therefore, it is necessary to use a one-step method, with the same order of accuracy, to compute enough starting values of the solution to be able to use the multistep method.

For example, to use the three-step Adams-Bashforth method, it is necessary to first use a one-step method such as the fourth-order Runge-Kutta method to compute y_1 and y_2, and then the Adams-Bashforth method can be used to compute y_3 using y_2, y_1 and y_0. Similarly, we sometimes use Runge-Kutta RK4 to evaluate $\widetilde{y}_{i+1}$ in the predictor-corrector method.

■ Question 65.

□

Check that the AM method for $s = 1$ is the Trapezoid rule. Then using the predictor-corrector method it becomes the improved Euler's method.